New Era University
COLLEGE OF INFORMATICS AND COMPUTING STUDIES
No. 9 Central Avenue, New Era, Quezon City
Tel. No: (02) 8981-4221 to 4231 loc3815

INTERNET OF THINGS

SeatSense: An IoT-Based Smart Public Seating Availability and Reservation System

SECTION: 4BSIT-1
GROUP NO.: 2
LEADER: Matanguihan, Nicole Charlly A.
MEMBERS:

Felix, Denverlee
Magarro, Layne Chester R.
Mamamao, Adrian C.
Mangabat, Bryan Paolo M.
Manguerra, Aryana Kristina S.
Mendoza, Adam Riggs
Mendoza, Marc Janzel M.
Mercado, John Derick O.
Perez, Lloyd Christian E.
Red, Alen Gem D.
Remo, Jhunelle A.
Rivas, Joseph E.
Rivera, Arren Jay M.
Riveral, Jaymar B.
Semilla, Laica Mae C.
Tan, Sophia Francine F.

ENGR. JOEMAR A. SILAO
Instructor

AREA OF INTEREST

Choose only what applies; put an X mark beside the interest:

____ Food Safety	____ Disaster Mitigation
____ Security	__x__ **Entrepreneurship**
____ Traffic / Road Congestion	____ Agriculture
____ Health	____ Environment
____ Education	

ABSTRACT OF THE PROPOSAL PROJECT

Busy dining places often struggle with long customer lines, unmanaged tables, and people taking seats before ordering or holding tables for too long. To solve these everyday problems, this project introduces SeatSense, a smart restaurant seating system. The system combines a touch-screen self-service kiosk with small screens attached to each table, connected in real time through light web technology and QR codes.

When customers arrive, they place their order at the kiosk, enter their name, select an open table on a map of the floor, and pay. The kiosk then prints a receipt with a special QR code. The moment the order goes through, the screen on the chosen table updates to RESERVED and starts a 10-minute timer for the customer to sit down. When the customer reaches their table, they scan the receipt's QR code on the table's scanner. If it matches, the display changes to OCCUPIED and starts a 2-hour dining timer. Sitting at the wrong table triggers a light and sound alert. If the restaurant is full, guests can join a queue on the kiosk for tables that are almost finished. Once a party leaves, the status updates to NEEDS_CLEANING so staff can wipe down the table and reset it using a master staff card. This simple setup cuts down waiting times, keeps seating fair, and helps restaurants serve more guests smoothly.

PROBLEM TO BE ADDRESSED

In fast-paced restaurants, managing tables by hand often leads to unhappy customers, slow service, and lost sales. A major reason for this is a lack of order: customers frequently grab tables before paying, stay long after finishing their meal, or sit at the wrong tables, which confuses the kitchen and delays food delivery.

Traditional waitlist tickets or buzzers do not tell staff or customers which tables are actually open or how soon a table will free up. This causes crowded waiting areas and leaves guests feeling frustrated. On top of that, when a table is vacated, staff might not notice immediately, leaving clean tables sitting empty while waiting guests stand outside. Without an automated, connected system that links ordering kiosks, table displays, and staff tasks, restaurants struggle to keep table turnover moving efficiently and fairly.

OBJECTIVES OF THE STUDY

The general objective of this study is to develop SeatSense, an IoT-based smart public seating availability and reservation system for dining establishments.

Specifically, this study aims to:

1. Design and build a self-service ordering kiosk using a Raspberry Pi 4 that lets customers browse the menu, select an available table from a live floor map, and print a receipt with a secure QR code.

2. Develop smart table hardware nodes using ESP32 microcontrollers equipped with QR scanners and OLED screens to display real-time seating statuses (VACANT, RESERVED, OCCUPIED, NEEDS_CLEANING).

3. Establish low-latency wireless communication using the MQTT protocol to sync table state changes instantly between the main server, kiosk, and table displays.

4. Implement automated check-in and seating enforcement by validating receipt QR codes at the table, triggering alerts if guests sit at the wrong table.

5. Create an automated queue and staff cleanup workflow that tracks dining timers, alerts staff when tables need cleaning, and lets them reset tables using a Master Staff QR Card.

PROPOSED TECHNOLOGY/SOLUTION TO THE PROBLEM

To fix the hassles of manual seating, long lines, and poor table visibility, this project introduces **SeatSense**—a smart seating and reservation system powered by IoT technology. Instead of relying on slow manual updates, SeatSense connects every part of the restaurant instantly using a lightweight messaging protocol called MQTT. The hardware setup includes a **Raspberry Pi 4-powered touchscreen kiosk** for ordering, **ESP32 microcontrollers** installed at each dining table, and a central **Node.js/Python backend server** that handles data processing.

At the core of the system is a central state manager running on the backend server. It enforces four primary table conditions: **VACANT**, **RESERVED**, **OCCUPIED**, and **NEEDS_CLEANING**.

1. **Ordering & Reservation:** When a customer orders at the kiosk, picks an open table on the interactive map, and pays, the kiosk prints a receipt with a secure QR code containing their order details and table assignment. Simultaneously, the server updates the table's OLED screen to **RESERVED** and starts a 10-minute arrival timer.

2. **Self Check-In & Enforcement:** Upon reaching the table, the customer holds their receipt QR code up to the table's scanner. If the code matches, the screen shifts to **OCCUPIED**, shows a greeting, and starts a 2-hour meal timer. If someone scans the wrong table, the system sounds an error alert to prevent seat hogging.

3. **Queueing & Cleanup:** When the restaurant is full, guests can pre-queue at the kiosk for tables that are near the end of their meal timers. Once a party leaves or their timer runs out, the table display changes to **NEEDS_CLEANING**. Staff members wipe down the space and

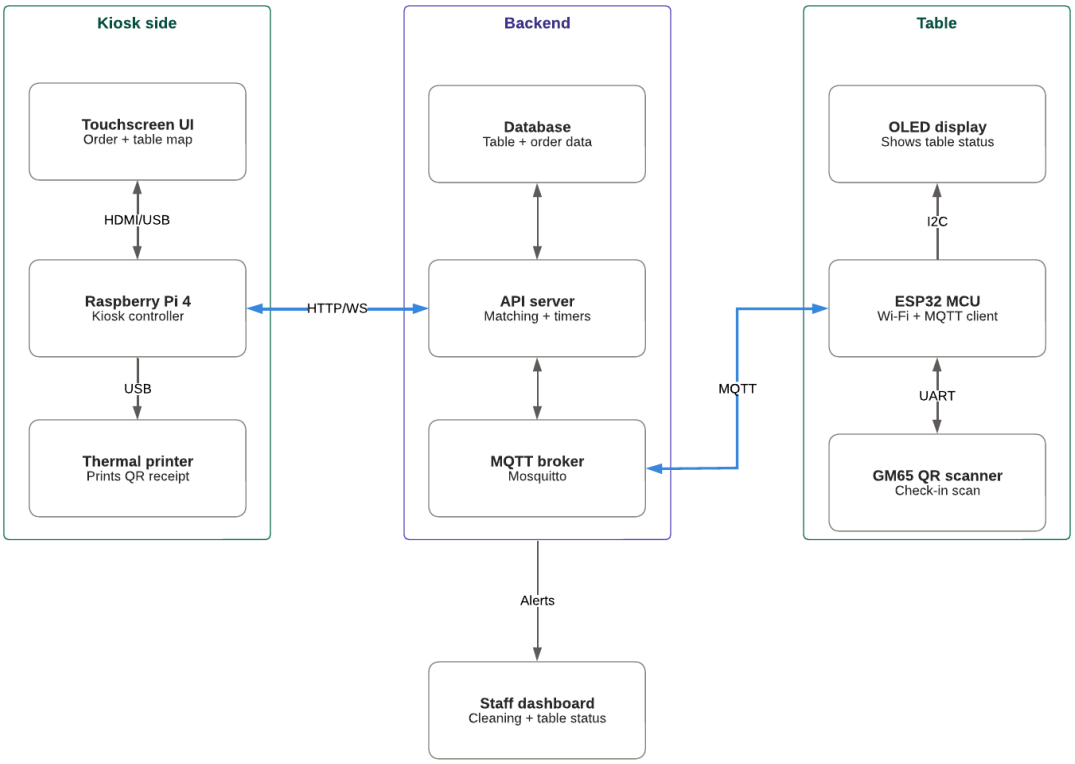
scan their Master Staff QR Card to mark the table **VACANT** (or reserve it automatically for the next waiting party).

SYSTEM DESIGN AND TECHNICAL DELIVERABLES

The following sections detail the structural, functional, visual, and physical components of the proposed system deliverables. Each section outlines the specific architectural design and operational role of the respective component.

1. BLOCK DIAGRAM

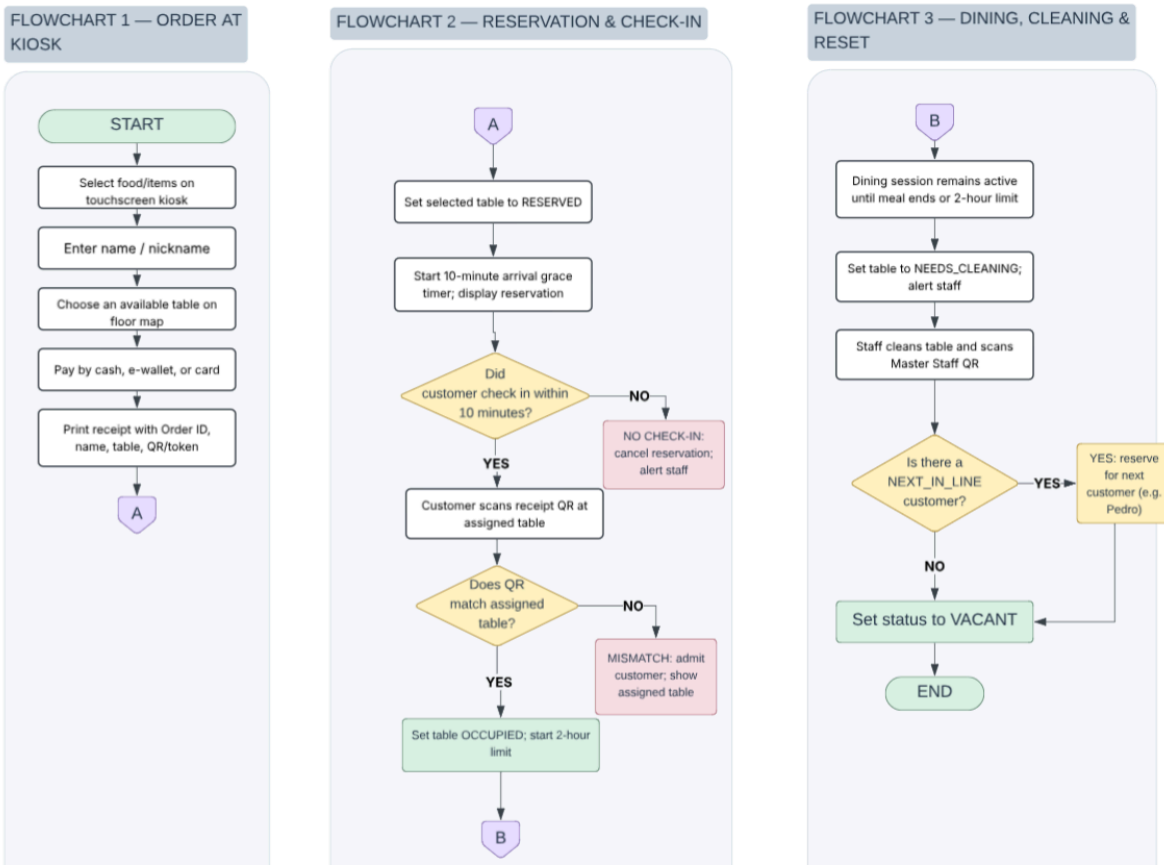
The System Block Diagram illustrates the high-level structural breakdown and data communication lines across the physical hardware components and backend services. Hardware inputs from the touchscreen monitor and thermal printer interact directly with the Raspberry Pi 4 Kiosk, which transmits order payloads to the backend via HTTP REST APIs or WebSockets.



Simultaneously, the backend communicates with the Eclipse Mosquitto MQTT Broker to publish lightweight, real-time command payloads over Wi-Fi to individual ESP32 table microcontrollers, driving local 1.3" OLED displays, reading GM65 QR scanner inputs via 4-channel logic level converters, and managing battery-powered step-up power circuits.

2. FLOWCHART

The System Flowchart maps the complete operational state transitions and conditional logic executed across the 5-step customer dining lifecycle. Beginning at Kiosk order placement, the logic branches through table selection, database FSM state updating to RESERVED, and receipt printing.

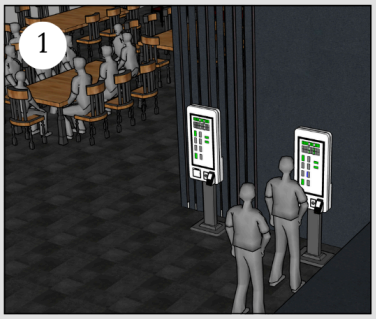


It then details the validation logic executed when scanning the QR code at the table module—verifying a match to shift the state to OCCUPIED versus signaling a mismatch warning—followed by the 2-hour session countdown timer, automated transition to NEEDS_CLEANING, and final reset via the staff's Master Staff QR Card.

2.2 STORYBOARD OF SYSTEM OPERATION

The storyboard illustrates the step-by-step interaction between customers, the SeatSense self-service kiosk, smart table modules, and restaurant staff. It presents the complete customer dining journey, from ordering and table reservation to QR-based check-in, dining session monitoring, and table cleaning. The storyboard provides a visual representation of how the proposed system operates in an actual restaurant environment.

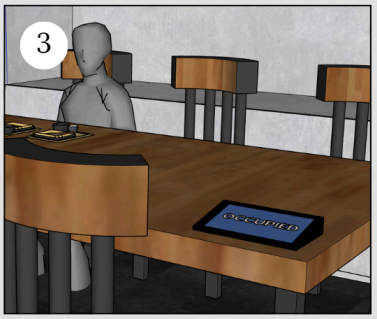
SEATSENSE STORYBOARD



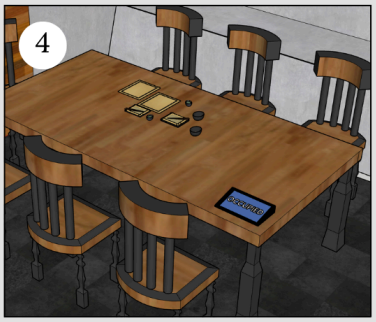
The customer enters the restaurant and proceeds to the SeatSense kiosk to begin the table selection process.



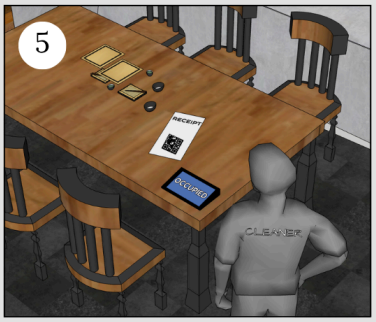
The customer uses the kiosk to view the available tables and selects a preferred table.



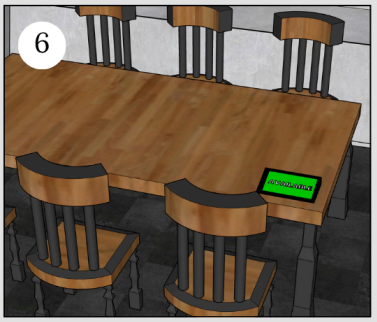
After the customer selects a table, the SeatSense system automatically marks it as OCCUPIED, and the mini screen on the selected table displays its updated status.



The customer enjoys their meal at the selected table. After eating, the customer leaves the restaurant, but the table remains marked as OCCUPIED because it has not yet been checked by the cleaner.



The cleaner approaches the empty table and scans its QR code using a mobile device to confirm that the table is ready for cleaning and reuse.



After the QR code is scanned, the SeatSense system updates the table status to AVAILABLE. The kiosk display and the table's mini screen are also updated to reflect the new status, allowing other customers to select the table.

3. TABLE OF HARDWARE AND SOFTWARE TOOLS

The Hardware and Software Tools deliverable itemizes the technological ecosystem powering both the physical hardware nodes and the supporting software stack. The Kiosk hardware consists of a Raspberry Pi 4 Model B (4GB RAM) running Chromium in Kiosk Mode, paired with a 15.6"/21.5" Full HD Touchscreen, a USB Thermal Printer with a dedicated power adapter, and a SanDisk 32GB Class 10 MicroSD card.

[INSERT TABLE: Hardware & Software Table]

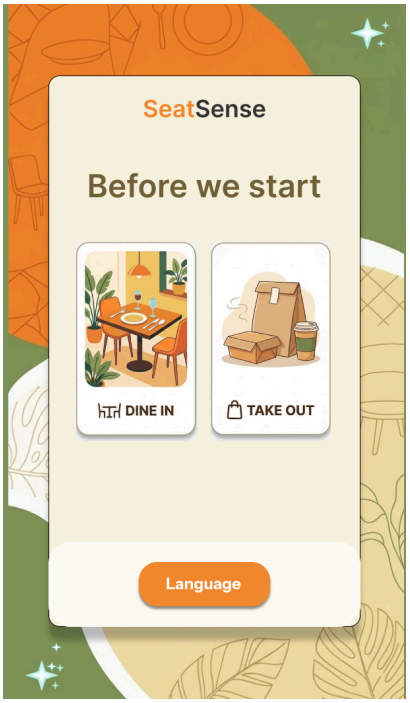
The Smart Table node is driven by an ESP32-WROOM-32 microcontroller connected to a GM65 QR Scanner Module via a 3.3V-to-5V Logic Level Converter, a 1.3" I2C OLED Display, two 18650 Li-ion batteries with a TP4056 charging module, and an MT3608 boost converter. The software stack uses React.js/Vue.js/Electron for the Kiosk UI, Node.js/Python for backend logic, MySQL/PostgreSQL/Firebase for DBMS, Mosquitto for MQTT messaging, and Arduino IDE/PlatformIO with Adafruit and PubSubClient libraries for the ESP32 C++ firmware.

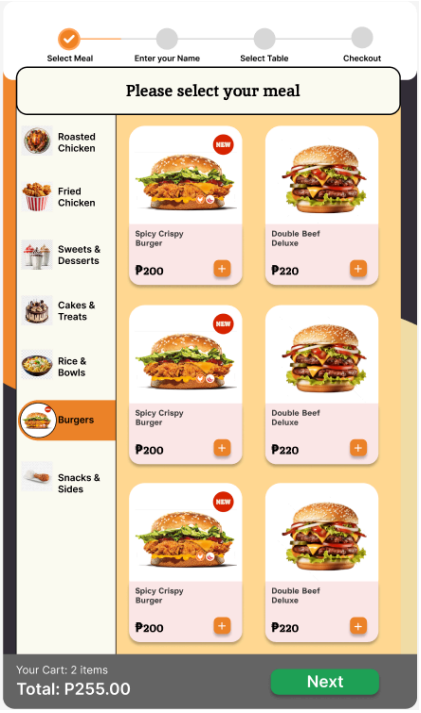
4. KIOSK INTERFACE

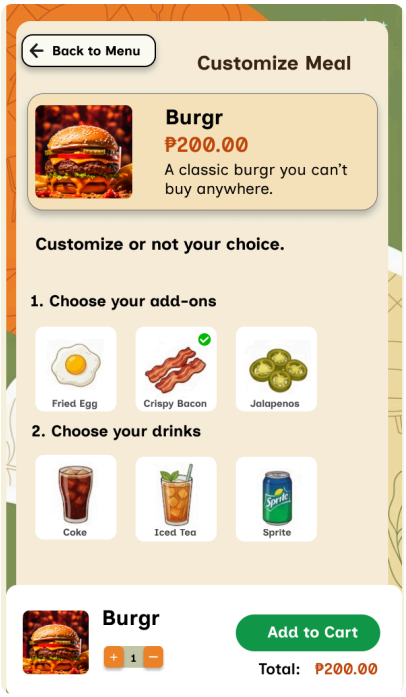
The Kiosk Interface serves as the primary customer self-service entry point, featuring a full-screen interactive ordering application designed to minimize check-in friction. The interface opens with an intuitive menu selection screen, prompts the user with an "Enter your Name/Nickname:" modal, and opens an Interactive Floor Map highlighting real-time vacant versus occupied tables.

Upon payment completion, the UI communicates with local ESC/POS printer libraries to render a QR code containing a JSON payload (Order ID, Customer Name, Assigned Table, and Hash Token) before returning to the welcome screen.

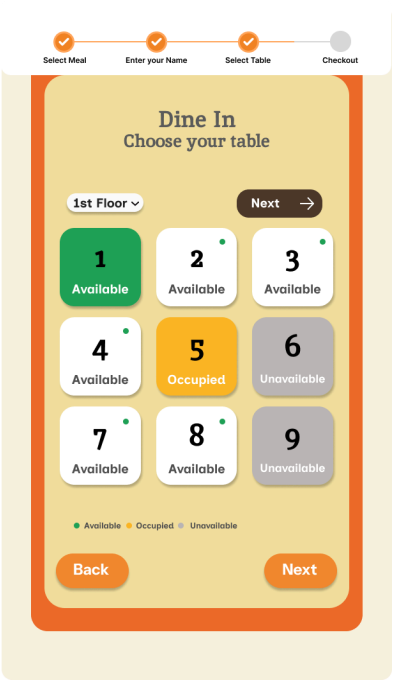




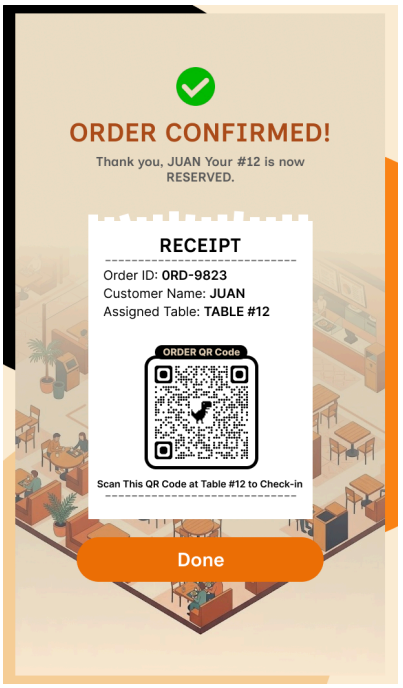



Back to Menu
Customize Meal
Burgr
₱200.00
A classic burgr you can't buy anywhere.
Customize or not your choice.
1. Choose your add-ons
Fried Egg
Crispy Bacon
Jalapenos
2. Choose your drinks
Coke
Iced Tea
Sprite
Burgr
Add to Cart
Total: ₱200.00


Select Meal
Enter your Name
Select Table
Checkout
SeatSense
ENTER YOUR NAME
What should we call you?
Type Nickname here..
Back
Choose Table


Select Meal
Enter your Name
Select Table
Checkout
Dine In
Choose your table
1st Floor
Next
1 Available
2 Available
3 Available
4 Available
5 Occupied
6 Unavailable
7 Available
8 Available
9 Unavailable
Available Occupied Unavailable
Back
Next

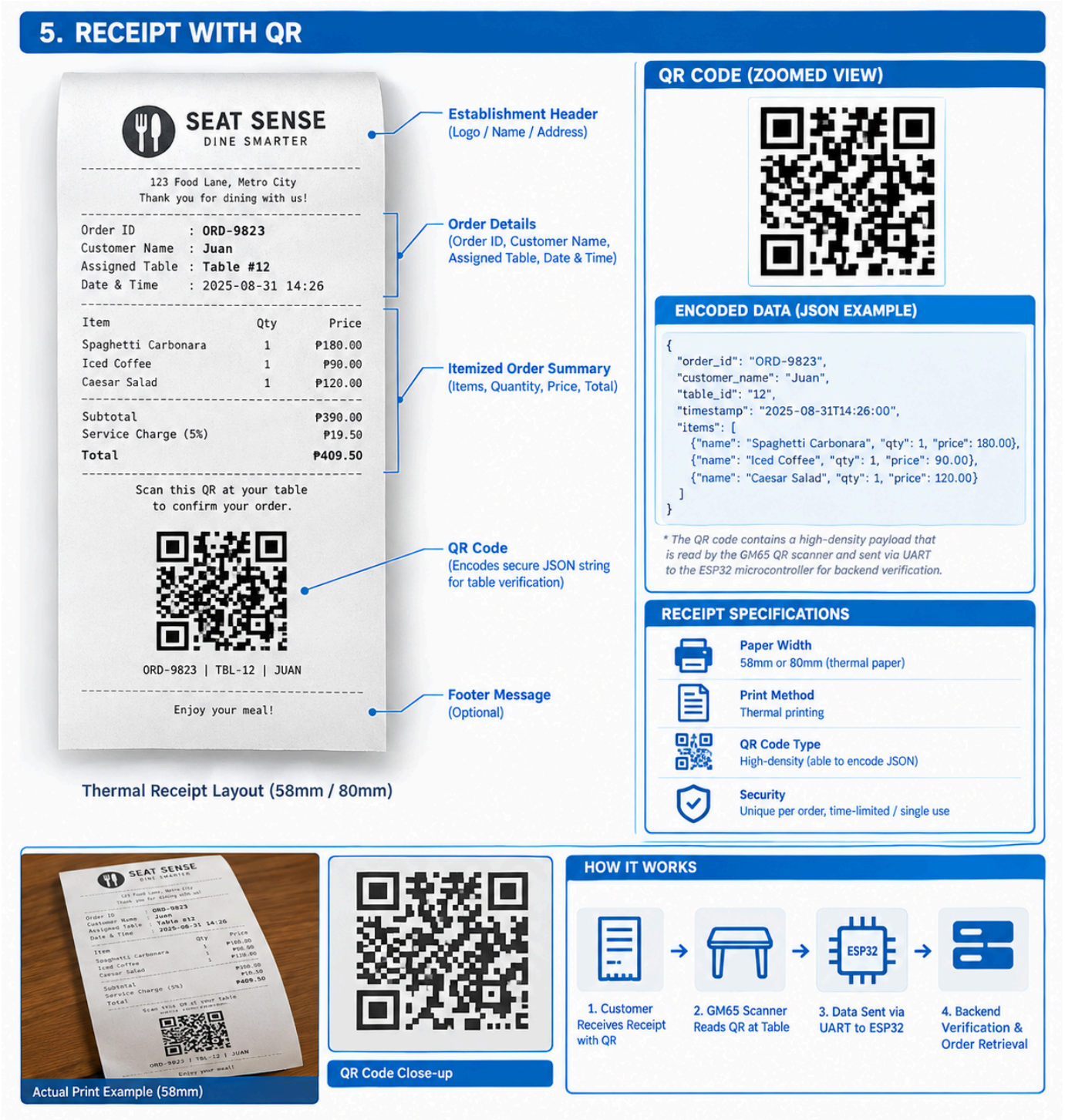

Select Meal
Enter your Name
Select Table
Checkout
ORDER SUMMARY
NAME: JUAN
Table Reserved: Table #1 (4-seater)
1x Tapsilog ₱120.00
1x Pork Sisig Rice ₱135.00
TOTAL: ₱255.00
Select Payment Method
Gcash
Cash
Back


ORDER CONFIRMED!
Thank you, JUAN Your #12 is now RESERVED.
RECEIPT
Order ID: ORD-9823
Customer Name: JUAN
Assigned Table: TABLE #12
ORDER QR Code
Scan This QR Code at Table #12 to Check-in
Done

5. RECEIPT WITH QR

The Receipt with QR acts as the physical key connecting the customer's order to their assigned dining station. Printed on 58mm or 80mm thermal paper, the receipt prominently displays the establishment header, Order ID (e.g., ORD-9823), Customer Name (Juan), Assigned Table (Table #12), and itemized order summary.

[INSERT IMAGE: Thermal Receipt Layout featuring Order Details, Assigned Table Identifier, and Unique High-Density Payload QR Code]

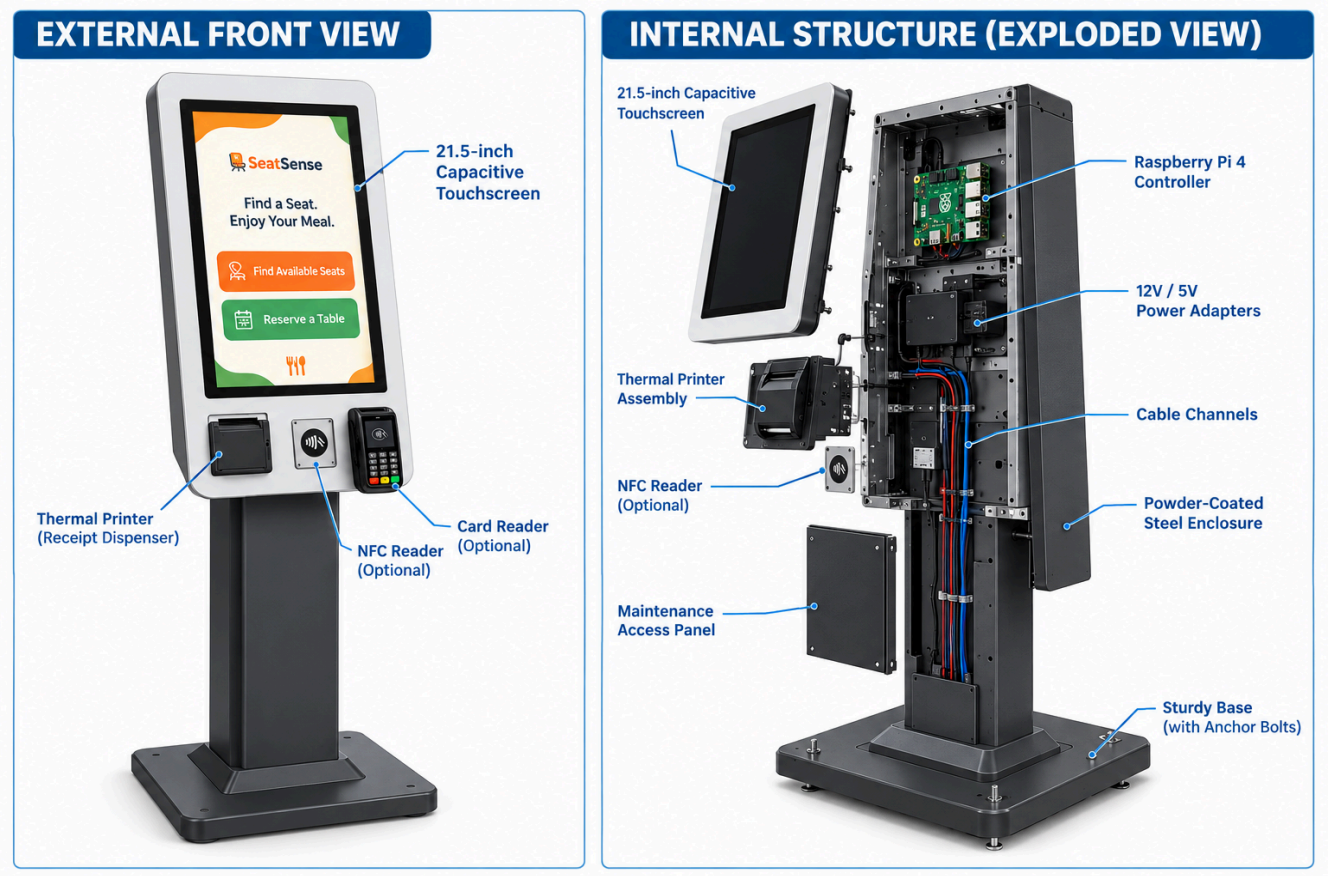


At the center of the receipt is a high-density printed QR code encoding the secure JSON string, allowing the GM65 QR scanner at the assigned table to quickly parse UART data into the ESP32 microcontroller for instantaneous backend verification.

6. KIOSK EXTERNAL AND INTERNAL LOOK WITH PARTS

This deliverable defines the ergonomic structure, physical enclosure, and internal component layout of the floor-standing self-service terminal. Built from durable powder-coated sheet metal, the outer chassis features a flush-mounted 15.6"/21.5" capacitive touchscreen display angled for comfortable viewing and cutouts for the thermal printer dispenser.

[INSERT IMAGE: Kiosk External Front View and Internal Structural Exploded View showing Raspberry Pi 4, Power Adapters, Printer Assembly, and Cable Channels]



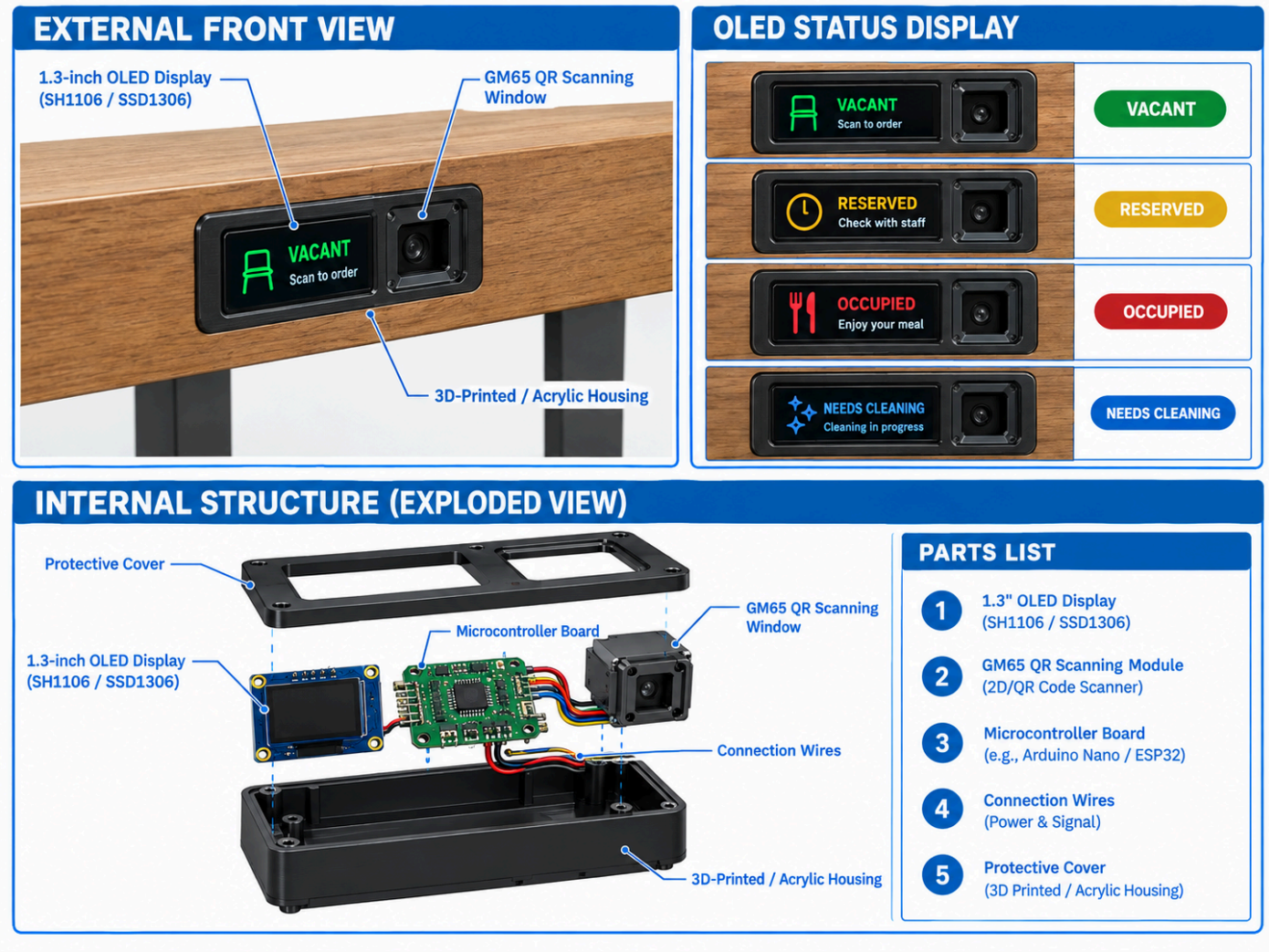
Internally, an official 5V 3A USB-C power supply powers the Raspberry Pi 4 Model B, while a separate dedicated 12V/24V high-current power adapter feeds the thermal receipt printer module. Micro-HDMI and USB touch cables connect the Pi 4 to the screen, with organized wiring conduits isolating high-voltage power lines from signal cables.

7. TABLE EXTERNAL AND INTERNAL LOOK WITH PARTS

This section details the physical styling, visual displays, and internal electronic assembly integrated at each smart dining station. Enclosed in a compact 3D printed or acrylic housing designed to protect against liquid spills, the external face features the 1.3" SH1106/SSD1306 OLED display screen showing real-time status text (VACANT, RESERVED, OCCUPIED, NEEDS_CLEANING) alongside the GM65 QR scanning window.

[INSERT IMAGE: Smart Table Module External View and Internal PCB Wiring Diagram

showing ESP32, Logic Level Converter, 18650 Battery Pack, TP4056, and MT3608 Modules]



Inside the enclosure, the ESP32 connects to the GM65 QR Scanner via a 4-channel Logic Level Converter to protect the 3.3V GPIO pins from 5V signals. Power is supplied by two 18650 rechargeable 3.7V Li-ion batteries through a TP4056 charging module with overcharge protection, boosted to 5V DC via an MT3608 step-up module to drive the scanner, display, and processor reliably without wall outlets.

REFERENCES

Atayero, A. A., Ilori, O. A., & Adedokun, M. O. (2015). Development of SeatSense: A wireless sensor network based seat detection system. In *Proceedings of the World Congress on Engineering and Computer Science 2015* (Vol. II, pp. 711–714). Newswood Limited. https://www.iaeng.org/publication/WCECS2015/WCECS2015_pp711-714.pdf

Breen, M., Dundon, M., & McCaffrey, C. (2018). Making every seat count: Space management at peak times in a university library. *New Review of Academic Librarianship*, 24(1), 105–118. https://doi.org/10.1080/13614533.2017.1414066

Geaga, H. A. J., & Satuito, A. B. (2026). *A comparative study on customers' experience between counter service and self-ordering kiosks in fast food restaurants* [Bachelor's thesis, University of the Philippines Diliman]. UP Diliman Digital Archives. https://digitalarchives.upd.edu.ph/item/63488/971/

Grunfeld, P. (2012). *Seat occupancy detection and display system* (U.S. Patent Application Publication No. 2012/0242492 A1). U.S. Patent and Trademark Office. https://patents.google.com/patent/US20120242492A1/en

Guzelgunler, Y. (2024). *Methods and systems optimizing swimming pool and restaurant service operations by utilizing occupancy detection device* (U.S. Patent No. 11,948,444 B2). U.S. Patent and Trademark Office. https://patents.google.com/patent/US11948444B2/en

Ishii, S. (2022). *Information processing device, information processing method, recording medium, and seat reservation system* (U.S. Patent No. 11,526,815 B2). U.S. Patent and Trademark Office. https://patents.google.com/patent/US11526815B2

Mahadevaswamy, U. B., & Gurudutt, M. S. (2019). IoT based occupancy detection using wireless controller in a building/workspace. *International Journal of Innovative Technology and Exploring Engineering*, 8(10), 609–614. https://www.ijitee.org/wp-content/uploads/papers/v8i10/I8947078919.pdf

Martínez-Sala, A. S., Hernando-Cánovas, L., Sánchez-Aarnoutse, J. C., & Alcaraz, J. J. (2025). Resource-efficient fog computing vision system for occupancy monitoring: A real-world deployment in university libraries. *Internet of Things*, 34, Article 101748. https://doi.org/10.1016/j.iot.2025.101748

Oh, C.-S., Jeong, S.-H., Kim, Y.-H., & Kim, T.-Y. (2016). Internet of Things based cafe seat confirmation system of implementation. *Proceedings of the Korean Institute of Information and Communication Sciences Conference*, 2016(10a), 593–595. https://koreascience.or.kr/article/CFKO201623965830490.page

Prithivirajan, M., & Shim, K. J. (2019). An IoT-driven smart cafe solution for human traffic management. In *2019 IEEE International Conference on Big Data (Big Data)* (pp. 6186–6188). IEEE. https://doi.org/10.1109/BigData47090.2019.9006188

Raj, A. (2024, July 26). *QR code scanner API for low-power embedded SoC boards*. Circuit Digest. https://circuitdigest.com/article/qr-code-scanner-api-for-low-power-embedded-soc-boards

Restaurant table management system (European Patent Application Publication No. EP 1 581 850 A2). (2005). European Patent Office. https://data.epo.org/gpi/EP1581850A2

Stambaugh, R. W. (2005). *Restaurant table turn system network* (U.S. Patent No. 6,980,088 B2). U.S. Patent and Trademark Office. https://patents.google.com/patent/US6980088B2/en

Thummar, D. P., Upadhyay, J. B., & Satote, B. R. (2025). DineWise RMS: A QR code-based restaurant management system for table booking and queue management. *International Research Journal on Advanced Science Hub*, 7(10), 865–872. https://doi.org/10.47392/IRJASH.2025.095

Wang, Q., Patel, H., & Shao, L. (2023). A longitudinal study of the occupancy patterns of a university library building using thermal imaging analysis. *Intelligent Buildings International*, 15(2), 62–77. https://doi.org/10.1080/17508975.2022.2147129

Hwang, J. (2008). Restaurant table management to reduce customer waiting times. *Journal of Foodservice Business Research*, 11(4), 334–351.

https://doi.org/10.1080/15378020802414612

Guzelgunler, Y. (2024). *Methods and systems optimizing swimming pool and restaurant service operations by utilizing occupancy detection device* (U.S. Patent No. 11,948,444 B2). U.S. Patent and Trademark Office.
https://patents.google.com/patent/US11948444B2/en

ESTIMATED PROJECT COST

Part of creating a sound proposal is to properly account the budget that your project would cost. From materials to labor costs, please include the possible bill of materials that the project will utilize.

No.	Quantity	Item Description	Unit	Unit Price	Subtotal
1	1	Raspberry Pi 4 Model B (4GB RAM)	Pc	₱4,200	₱4,200
2	1	Official 5V 3A USB-C Power Supply	Pc	₱650	₱650
3		SanDisk Extreme 32GB MicroSD Card (Class 10)	Pc	₱450	₱450
4	1	Touchscreen Display Monitor (15.6" o 21.5" Full HD)	Pc	₱3,500	₱3,500
5	1	Micro-HDMI to Standard HDMI Cable + USB Cable	Set	₱350	₱350
6	1	Thermal Receipt Printer Dispenser (58mm / 80mm USB Thermal Printer)	Pc	₱2,200	₱2,200
7	1	12V / 24V Dedicated Power Adapter (para sa Thermal Printer)	Pc	**₱350**	**₱350**
8	1	Thermal Paper Roll (58mm o 80mm)	Roll	₱2,200	₱2,200
9	1	Wi-Fi Module / Ethernet Port (Built-in sa Pi 4)	Set	₱350	₱350
10	1	Kiosk Metal Enclosure / Casing	Set	₱35	₱35
11	1	SanDisk Extreme 32GB MicroSD	Pc	₱0 *(Built-in)*	₱0

		Card (Class 10)			
12	1	Touchscreen Display Monitor (15.6" o 21.5" Full HD)	Pc	₱3,500	₱3,500
13	1	Micro-HDMI to Standard HDMI Cable + USB Cable	Pc	₱450	₱450
14	1	ESP32 Development Board (ESP32-WROOM-32)	Pc	₱3,500	₱3,500
15	1	GM65 / MH-ET LIVE Barcode & QR Code Scanner Module	Pc	₱350	₱350
16	1	1.3" I2C OLED Display Module (SH1106 / SSD1306)	Pc	₱250	₱250
17	1	4-Channel Logic Level Converter (3.3V to 5V)	Pc	₱1,200	₱1,200
18	2	18650 Rechargeable Li-ion Batteries (3.7V 2600mAh - 2 Pcs)	Pcs	₱50	₱50
19	1	TP4056 Micro-USB / Type-C Charging Module	Pc	₱150	₱300
20	1	MT3608 Step-Up Boost Converter Module	Pc	₱30	₱30
21	1	Jumper Wires & Breadboard / PCB Board	Pc	₱150	₱150
22	1	3D Printed / Acrylic Table Enclosure	Pc	₱1,200	₱1,200
				Grand Total	25,480

PATENTABILITY CHECKING

Patent Searching is a technique to check whether your product, idea, or project has already been invented and patented by a certain individual, groups, or companies.

Steps:

1. Determine technical search keywords that your project proposal would possibly fall in.
2. Go to patent search websites such as WIPO's PATENTSCOPE (https://patentscope.wipo.int/search/en/search.jsf), Google's Patents (https://patents.google.com/), Lens (lens.org/lens/search/patent/structured), Espacenet (https://worldwide.espacenet.com/), US Patent and Trademark Office (https://www.uspto.gov/patents/search) , and IPO Philippines (https://www.ipophil.gov.ph/patent/).
3. Use these websites to be able to search for similar projects, research, and patents using keywords you have developed.
4. List certain projects that are similar in nature but can be different in implementation. Please provide the following details:

No.	Project/Research Title	Problem Statement	Key Similarities	Key Differences
1	*Methods and Systems Optimizing Swimming Pool and Restaurant Service Operations by Utilizing Occupancy Detection Device* (U.S. Patent No. 11,948,444 B2)	Restaurants and similar venues need a practical way to communicate table status and coordinate service.	Includes table-status indicators, user inputs, wireless communication, and reserved-status functionality.	SeatSense focuses on public seating availability and temporary absence management, with a proposed sensor, database, and dashboard integration. The extent of overlap must be evaluated against the complete patent claims.
2	Restaurant Table Management System (EP 1581850 A2; US 2004/0138929 A1)	Restaurants need to know right away which tables are vacant and ready for the next party, without checking manually.	A computer displays each table as an icon with a status of vacant, occupied, or anticipated to be vacant soon. When a table's occupancy duration exceeds a threshold, the system infers that the party will likely leave soon. This is similar to SeatSense's dining timer and automatic pre-queue.	The system uses a switch and wireless transmitter at each table to signal vacancy, and a table can also be returned to vacant by a manual command at the computer. It has no self-service kiosk, QR verification, RESERVED state with a grace period, NEEDS_CLEANING workflow, MQTT synchronization, or automated FIFO pre-queue

...	US11526815B2 (application US 2019/0057325), "Information Processing Device, Information Processing Method, Recording Medium, and Seat Reservation System,"	Customers struggle to find and secure available seating in crowded venues, while management lacks automated tools to track real-time occupancy and predict when occupied tables will become vacant for waiting customers.	The patented system by Ishii (2022) shares several core functional concepts with SeatSense regarding automated dining management. Both systems streamline the customer journey by coupling seat selection directly with the payment and order placement process. Additionally, both implementations provide live, real-time seat occupancy updates to help customers find available tables. Finally, Ishii (2022) incorporates predictive queueing that estimates future table vacancy times using initial occupancy timestamps, which aligns closely with SeatSense's pre-queueing mechanism for tables near the end of their dining session.	Despite their operational similarities, SeatSense differs significantly from Ishii (2022) in its hardware design, seating verification, and operational state workflow. While Ishii (2022) relies on overhead camera image processing to detect vacant tables, SeatSense utilizes physical, battery-powered table hardware nodes equipped with OLED screens and QR scanners. Unlike Ishii's system, SeatSense incorporates a 10-minute arrival grace period upon order payment, table-side QR scan matching to enforce correct seating and prevent seat-hogging, and an explicit needs_cleaning state that requires a Master Staff QR Card to reset tables. Furthermore, SeatSense establishes a lightweight, WebSocket- and MQTT-synchronized state machine to manage real-time communication across kiosks, backend servers, and table displays

3.	Seat Occupancy Detection and Display System (US Patent App. 2012/0242492 A1)	Venues and transportation systems need to efficiently ascertain if a specific seat is occupied without manual physical checks.	Employs a seat occupancy sensor array under the cushion to detect weight-bearing status.	Designed for arena/airplane ticketing and remote occupancy mapping via a central processor, not for monitoring individual ergonomic posture.
4.	Methods and Systems Optimizing Service Operations via Occupancy Detection (U.S. Patent No. 11,948,444 B2) Guzelgunler, 2024	Restaurants and recreation facilities lack an integrated way to monitor table status, communicate availability, and coordinate service. Manual updates cause delays, inconsistent information between staff and guests, and inefficient table use during peak hours. No unified system connects occupancy tracking, reservation management, and customer notifications.	• Uses occupancy detection devices • Includes table-status indicators • Supports wireless communication • Covers restaurant service operations • References reservation-type functionality	• Broad scope includes swimming pools and general facilities → SeatSense designed exclusively for restaurant dining workflows • Detection method unspecified — hardware-oriented → SeatSense QR-first approach: low-cost, no extra hardware, customer-inclusive • No defined multi-state table lifecycle → SeatSense structured 4-stage workflow guides cleaning & readiness • No customer-facing standby system → SeatSense lets guests

				join queue remotely, track progress, receive alerts • No timing safeguards → SeatSense grace periods & dining timers actively manage turnover
5.	*Resource-Efficient Fog Computing Vision System for Occupancy Monitoring* (Martínez-Sala et al., 2025)	Public facilities require real-time, low-cost, continuous occupancy monitoring and people-flow tracking across shared spaces without high-cost hardware	Uses overhead camera vision monitoring, computer vision inference (YOLOv5), and fog computing to detect human presence and track occupancy	Martínez-Sala et al. focus on continuous passive tracking and entrance flow detection. SeatSense pairs camera feeds with customer receipt QR validation, an enforced 2-hour session timer, a "Needs Cleaning" state, and master QR custodial verification
6.				

Note: Add additional rows if necessary.

REVIEW OF RELATED WORKS

This section requires you to highlight work done by others that somehow ties in with your own work. It, basically, may be work that you're basing your work off, or work that shows the attempts of others to solve the same problem. Similar steps from patent searching is done to identify related research and projects. Please provide the following details:

No.	Project/Research Title	Problem Statement	Methods or Technology Used	Strengths	Weaknesses
1.	An IoT-Driven Smart Cafe Solution for Human Traffic Management (Prithivirajan & Shim, 2019)	Corporate staff would go down to a cafe only to find that it had no free seats or tables,	Ultrasonic sensors monitor seat occupancy and Bluetooth Low Energy beacons	Lets people check seat and table availability in near real time before going to the cafe.	Only a preliminary test on a few tables. The first sensor layout could not monitor about 48%

		and the owner could not tell how busy the cafe was.	track movement. Raspberry Pi and GrovePi units send the data to a central database on Amazon Web Services through a Node-RED pipeline, and the results can be viewed in a web browser.	Uses real sensors and was tested on a few tables to find the best detection distance. Explains why sensors were chosen over point-of-sale data, Wi-Fi data, and cameras (speed and privacy). Helps the owner plan the floor layout.	of the seats (72% after a revision). Sensors are wired to Raspberry Pi units that need power sockets. It focuses on showing occupancy, with queue management planned only for later. No mention of ordering, QR check-in, or prevention of seat squatting. The last point is inferred from the paper and not confirmed.

2.	A Comparative Study on Customers' Experience Between Counter Service and Self-Ordering Kiosks in Fast Food Restaurants (Geaga & Satuito, 2026)	Fast-food restaurants face operational bottlenecks and customer dissatisfaction across service touchpoints (queueing, ordering, and order pickup), but local evidence comparing customer experiences and operational efficiency between traditional counter service and self-ordering kiosks in Metro Manila remains limited.	The study evaluated customer experiences with self-ordering kiosks versus traditional counter service across three key touchpoints: queueing, order placement, and order acquisition. Using a Qualtrics-based online survey, the authors gathered data from Metro Manila respondents aged 18 to 59 who had experience using both ordering systems.	Provides valuable localized data showing that self-ordering kiosks effectively reduce total time spent in fast-food establishments. and it demonstrates that overall customer satisfaction with kiosk technology significantly influences their intention to reuse self-service systems.	The research highlights customer-reported concerns regarding kiosk order accuracy. Additionally, because it focused strictly on front-of-house ordering mechanics as an undergraduate thesis, the study did not account for broader restaurant operations such as table allocation, real-time seating availability, or IoT hardware integration.
3.	Development of SeatSense: A Wireless Sensor Network Based Seat Detection System (Atayero et al., 2015)	Finding available seats in large auditoriums and similar venues is difficult without centralized occupancy information.	Per-seat occupancy sensors, low-power microcontrollers, wireless sensor nodes, and a central base station.	Provides a seat-level monitoring architecture with centralized visualization and event-based status reporting.	Focuses on occupancy detection rather than user-controlled temporary absence, reservation management, or the proposed SeatSense workflow.
4.	Real-time Sitting Correction Using Arduino-Based Ultrasonic Sensor at the	Prolonged improper sitting at workstations causes chronic	Arduino Uno microcontroller connected to HC-SR04	Achieves a high overall detection accuracy (97.4%) using highly	Relies entirely on distance measurement (ultrasonic) rather

	Workplace (Jameel, 2024)	spinal issues that users are unaware of in real time.	ultrasonic sensors to measure the user's backrest proximity.	affordable, accessible hardware.	than mapping actual physical weight distribution (load cells) on the seat cushion.
5.	Virtual Waiting System Quality and Customer Satisfaction in Restaurants (Kim et al., 2021)	Traditional physical waiting lines create frustration and perceived unfairness. Customers cannot easily see their position or estimate wait duration, leading to anxiety, early departures, and negative impressions of service quality. Most existing systems do not provide	Collected survey responses from 385 restaurant patrons; used Partial Least Squares Structural Equation Modelling (PLS-SEM) to analyze how virtual queue design factors transparency, fairness, information quality impact customer	Confirmed that digital virtual queues with real-time position updates and estimated wait times significantly improve satisfaction, perceived fairness, and willingness to return. Proved that transparent communication alone can make waiting feel shorter and more acceptable to	Validated only the benefits of virtual queuing did not integrate table occupancy tracking, reservation management, or cleaning status workflows. No multi-state table lifecycle, no grace periods for late arrivals, no dining duration timers, and no unified platform connecting queue

		transparent, real-time information leaving patrons uncertain and less satisfied even when wait times are reasonable.	satisfaction and revisit intention.	customers validating the value of digital, customer-facing queue systems.	status to actual table availability.
6.	*Resource-Efficient Fog Computing Vision System for Occupancy Monitoring* (Martínez-Sala et al., 2025)	Inaccurate information about space availability makes facility management difficult in shared facilities. Public spaces require real-time, continuous occupancy monitoring without high hardware costs.	Ceiling-mounted, top-view cameras placed above entrances transmitting video over intranet to a single CPU-only fog server running a fine-tuned YOLOv5 model for batch processing.	Demonstrates high detection accuracy (94%) for tracking people flow using resource-efficient, low-cost commodity hardware in production over 12 months.	Focuses purely on continuous passive occupancy detection and entrance flow; lacks active operational controls like table cleaning verification, QR validation, or session enforcement.
Submitted Proposal					

Note: Add additional rows if necessary.

PROJECT SCHEDULE

Provide a tentative timetable that shows the start and end date of all project tasks using a Gantt Chart. This should reflect flexible working arrangements allowing remote work on the proposed project. Use the table below, add more rows if necessary, or attach your own Gantt chart.

SeatSense: An IoT-Based Smart Public Seating Availability and Reservation System

Start Date: August 2026
End Date: November 2026

